

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular End Semester Examination – Winter 2022 Course: B. Tech. Branch : Mechanical Semester :V Subject Code & Name:BTMOE505D(Product Design Engineering) Max Marks: 60 Date:14/02/2023 Duration: 3.00 Hr.		
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.		
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	Define the product. Explain the various reasons that trigger new product development.	CO1	6
B)	Write a short note on reverse engineering .What is the benefits and limitation of reverse engineering approach to product design.	CO1	6
C)	Discuss in brief the problem solving approach for Product Design.	CO1	6
Q.2	Solve Any Two of the following.		
A)	Define ideation. Explain any two techniques of ideation.	CO2	6
B)	Define product architecture. Differentiate between the integral and modular architecture.	CO2	6
C)	What is Market research? Explain the importance of Market research in Product Design Engineering.	CO2	6
Q. 3	Solve Any Two of the following.		
A)	Define prototype. What are different types of prototypes?	CO3	6
B)	New Product Development is task of Multidisciplinary team. Explain the statement in detail with an example.	CO1	6
C)	Discuss in brief the health and hazard related issues of process operations in industries with suitable examples.	CO3	6
Q.4	Solve Any Two of the following.		
A)	What is geometrical modelling? Explain wireframe, solid and surface modelling with sketches.	CO4	6
B)	Write a short note on CAD/CAM.	CO4	6
C)	What is the purpose of materials and safety data-sheet? What is the	CO4	6

	minimum content of the materials and safety data-sheet?		
Q. 5	Solve Any Two of the following.		
A)	Explain the Product Life Cycle management approach related to environmental concerns of a product.	CO5	6
B)	Write a short note on sustainable waste management.	CO5	6
C)	Explain in brief the concepts of recycle and reuse with a case study.	CO5	6
	*** End ***		

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